Quasiregular Reeb Dynamics on Pairwise-Coprime Brieskorn Three-Manifolds

HCS-C370 theorem package

4 September 2026

Abstract

For every pair of odd coprime integers $3 \le p < q$, we close the normalized Reeb dynamics of the Brieskorn link $\Sigma(2, p, q)$. There are exactly three exceptional simple orbit circles, of periods $2p, 2q, pq$, and a principal period $2pq$. A divisibility rule gives every fixed-time component and proves it is Morse–Bott. Missing-coordinate trivializations give the transverse rotations, return determinants, first degeneracies, and all pre-degeneracy Conley–Zehnder indices. The orbit space is $S^2(2, p, q)$, and a declared Milnor-fiber capping convention gives $\mu_{\mathrm{RS}} = 2(2pq)\chi_{\mathrm{orb}} = -2pq + 4p + 4q$; this is positive only for $(p, q) = (3, 5)$ and never zero. Exhaustive exact computation is a regression receipt, not the proof.

1 Frozen contact system and main result

Put $a = (a_0, a_1, a_2) = (2, p, q)$ and

$$f(z) = z_0^2 + z_1^p + z_2^q, \qquad \Sigma_{p,q} = f^{-1}(0) \cap S^5 \subset \mathbb{C}^3.$$

The origin is the only critical point of f, so $\Sigma_{p,q}$ is a smooth closed three-manifold. We freeze the scale

$$\alpha = \frac{i}{4\pi} \sum_{j=0}^{2} a_j (z_j d\bar{z}_j - \bar{z}_j dz_j).$$

This is the standard weighted Brieskorn contact form in the normalization used throughout this paper.

Theorem 1 (Complete Reeb atlas) *The Reeb vector field and its flow are*

$$R = 2\pi i \sum_{j=0}^{2} \frac{1}{a_j} (z_j \partial_{z_j} - \bar{z}_j \partial_{\bar{z}_j}), \qquad \Phi_t(z_j) = e^{2\pi i t / a_j} z_j.$$

There are three exceptional simple circles, with support and primitive period

$$\{0, 1\} : 2p, \qquad \{0, 2\} : 2q, \qquad \{1, 2\} : pq,$$

whereas the principal period is $d = 2pq$. For real T, let $J_T = \{j : T/a_j \in \mathbb{Z}\}$. The fixed set is empty for $|J_T| < 2$, the corresponding exceptional circle for $|J_T| = 2$, and all of $\Sigma_{p,q}$ for $|J_T| = 3$. Every nonempty component is Morse–Bott.

In the ambient missing-coordinate complex-line trivializations, the three transverse rotation numbers are

$$\rho_{01} = \frac{2p}{q}, \qquad \rho_{02} = \frac{2q}{p}, \qquad \rho_{12} = \frac{pq}{2}.$$

Thus $\det_{\mathbb{R}}(I - P) = 4\sin^2(\pi\rho)$, *and first degeneracy occurs at covers $q, p, 2$. Before that cover,*

$$\mu_{\mathrm{CZ}}(\gamma^r) = 2\lfloor r\rho \rfloor + 1.$$

The common-period orbit space is $S^2(2, p, q)$, with

$$\chi_{\mathrm{orb}} = \frac{1}{2} + \frac{1}{p} + \frac{1}{q} - 1.$$

In the standard Milnor-fiber capping trivialization, the principal family has

$$\mu_{\mathrm{RS}} = 2d\chi_{\mathrm{orb}} = -2pq + 4p + 4q.$$

Both quantities are positive exactly at $(p, q) = (3, 5)$, negative otherwise, and never zero.

2 Normalization and primitive orbit types

The weighted ambient two-form is positive, and its contact-type restriction to the link is $d\alpha$. Direct evaluation gives, coordinate by coordinate,

$$\alpha(R) = \sum_j |z_j|^2 = 1, \qquad \iota_R d\alpha = -d\|z\|^2, \qquad df(R) = 2\pi i f.$$

Therefore R is tangent to both defining hypersurfaces, and $\iota_R d\alpha$ vanishes on the tangent of the link. This proves the stated normalization rather than fixing it only up to a positive constant.

A link point cannot have only one nonzero coordinate. On a two-coordinate locus of weights a_i, a_j, the moduli are the unique positive solution of

$$r^{a_i} = s^{a_j}, \qquad r^2 + s^2 = 1.$$

The phase equation cuts out a connected one-dimensional subtorus because $\gcd(a_i, a_j) = 1$. Hence each such locus is one embedded circle and one Reeb orbit. A point of support S returns first at $\operatorname{lcm}(a_j : j \in S)$. Pairwise coprimality of $2, p, q$ now gives

support	01	02	12	012
period	$2p$	$2q$	pq	$2pq$
isotropy order	q	p	2	1
orbit-quotient dimension	0	0	0	2

The final entry is the structural obstruction to treating all primitive orbits as a discrete ledger.

3 All fixed times and the Morse–Bott kernel

A point z is fixed by Φ_T exactly when $e^{2\pi i T/a_j} = 1$ for every nonzero z_j. Since at least two coordinates are nonzero, this proves the asserted J_T classification for arbitrary real T, not just for the finite evidence window.

It remains to prove cleanness. Along an exceptional two-coordinate circle, varying the missing coordinate is tangent to $f^{-1}(0)$ to first order: the missing partial derivative $a_k z_k^{a_k - 1}$ vanishes at $z_k = 0$. It is also tangent to the sphere there. This complex line is the contact-normal plane to the Reeb orbit. At a time fixing exactly the active coordinates, the derivative is the identity on the orbit tangent and a nonidentity rotation on the missing line. Hence

$$\ker(d\Phi_T - I) = T\,\mathrm{Fix}(\Phi_T).$$

At a common-period time the derivative is the identity on the whole link, so the same equality holds. All nonempty fixed sets are therefore Morse–Bott.

For one pair, in the integer window $1 \le T \le 2pq$, the exact counts are

fixed class	$\varnothing$	01	02	12	$\Sigma_{p,q}$
count	$2pq - p - q$	$q - 1$	$p - 1$	1	1.

For example, the 01 times are the nonprincipal multiples of $2p$, hence $q - 1$ of them. The other entries follow identically, and their sum is $2pq$.

4 Transverse rotations, determinants, and CZ indices

The derivative on the missing k-coordinate line over an orbit of period T_0 is multiplication by $e^{2\pi i T_0/a_k}$. With the constant ambient coordinate vector as the declared complex trivialization, its rotation number is T_0/a_k. This yields $2p/q, 2q/p, pq/2$ without an unstated framing change.

For the real rotation matrix through angle $2\pi\rho$,

$$\det_{\mathbb{R}}(I - P) = 2 - 2\cos(2\pi\rho) = 4\sin^2(\pi\rho).$$

The fractions are reduced because p, q are odd and coprime. Their denominators $q, p, 2$ are exactly the first degenerate covers; multiplying each by its primitive period gives $2pq$. Before degeneracy the standard planar rotation formula is $\mu_{\mathrm{CZ}}(\gamma^r) = 2\lfloor r\rho \rfloor + 1$. We make no CZ assignment at the degenerate common-period cover; it belongs to the RS calculation below.